Accuracy vs two-qubit gate budget on held-out geometries

LiH (6q) — test (grid), 17 geometries

LiH (6q) — test_fresh, 14 geometries

H_4 (8q) — test (grid), 22 geometries

H_4 (8q) — test_fresh, 20 geometries

% within 1.6 mHa

median $|E - E_{FCI}|$ (mHa)

mean compiled CX count (Qiskit CX basis)

Each point = one budget level / prefix length (joined in level order), averaged over held-out geometries.
ADAPT needs a fresh ADAPT run per geometry; all other curves are one-shot (0 energy evaluations).

- budget-conditioned policy (one model, one-shot)
- budget policy before GRPO (budget SFT only)
- ADAPT truncated at k ops (fresh run, re-optimised; runs that stopped earlier drop out)
- angle KRR on ADAPT k-op prefixes (one-shot)
- angle KRR on pruned n-op labels (one-shot)
- angle KRR on canonical n-op prefixes (one-shot)
- SFT one-shot (fixed canonical sequence)